6.4 Solving Right Triangles

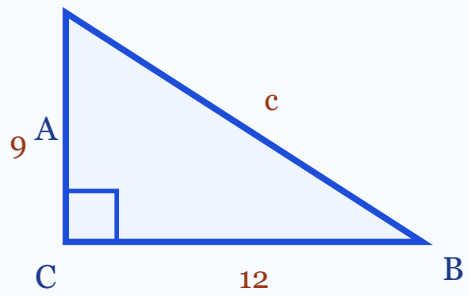
"Solving" a right triangle means finding every missing side and acute angle. Use the **Pythagorean Theorem** when two sides are known and you need the third. Use a **trig ratio** (SOH-CAH-TOA) when one side and one acute angle are known and you need another side. Use an **inverse trig function** ($\sin^{-1}$, $\cos^{-1}$, $\tan^{-1}$) when two sides are known and you need an angle — for example, if $\tan\theta = \frac{3}{4}$, then $\theta = \tan^{-1}(\frac{3}{4})$. Once one acute angle is known, the other follows immediately since the two acute angles are complementary ($90°$ minus the first). This worksheet combines all three tools, and several problems ask you to fully solve a triangle — finding all three sides and both acute angles.

Examples

Example 1: Two legs — find hypotenuse and an angle

Legs 9 and 12. Find the hypotenuse and both acute angles (express angles using inverse trig).

Solution



Two legs known; find c, then an angle.

Find the hypotenuse with the Pythagorean Theorem: $c^2 = 9^2 + 12^2 = 81 + 144 = 225$
$c = 15$.

Find one acute angle using tan: $\theta = \tan^{-1}(\frac{9}{12}) = \tan^{-1}(\frac{3}{4})$.

The other acute angle is the complement: $90° - \theta$.

Answer: $c = 15$; angles are $\tan^{-1}(\frac{3}{4})$ and $90° - \tan^{-1}(\frac{3}{4})$.

Example 2: Leg and angle — find the hypotenuse

One leg is 10, adjacent to a $35°$ angle. Find the hypotenuse.

Solution

Identify the ratio linking the adjacent side and the hypotenuse: CAH.

$$\cos 35° = \frac{10}{\text{hyp}}$$

Solve: $\text{hyp} = \dfrac{10}{\cos 35°}$.

Answer: $\text{hyp} = \dfrac{10}{\cos 35°}$.

Example 3: Hypotenuse and angle — find the opposite side

The hypotenuse is 20, and one acute angle is $40°$. Find the opposite side.

Solution

Identify the ratio linking the opposite side and the hypotenuse: SOH.

$$\sin 40° = \frac{\text{opp}}{20}$$

Solve: $\text{opp} = 20 \sin 40°$.

Answer: $\text{opp} = 20 \sin 40°$.

Example 4: Two sides — find the angle

Opposite side $= 5$, adjacent side $= 12$. Find the acute angle θ.

Solution

Identify the ratio linking opposite and adjacent: TOA.

$$\tan \theta = \frac{5}{12}$$

Solve using the inverse tangent: $\theta = \tan^{-1}\left(\dfrac{5}{12}\right)$.

Answer: $\theta = \tan^{-1}\left(\dfrac{5}{12}\right)$.

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Example 5: Verify a special triangle and name the angles

A triangle has sides 5, $5\sqrt{3}$, and 10. Verify it is a right triangle, then identify all three angles.

Solution

Check the Pythagorean Theorem with the longest side as the hypotenuse: $5^2 + (5\sqrt{3})^2 = 25 + 75 = 100 = 10^2$ ✓.

The side ratios $x : x\sqrt{3} : 2x$ (with $x = 5$) match the 30-60-90 pattern.

Answer: angles are $30°$, $60°$, and $90°$.

Example 6: Find the other acute angle

One acute angle of a right triangle is $52°$. Find the other.

Solution

The two acute angles of a right triangle are complementary: they sum to $90°$.

Other angle $= 90 - 52 = 38°$.

Answer: $38°$.

Example 7: Side then angle

Legs 8 and 15. Find the hypotenuse, then the angle opposite the leg of length 8.

Solution

Find the hypotenuse: $c^2 = 8^2 + 15^2 = 64 + 225 = 289$

$c = 17$.

Find the angle using SOH: $\theta = \sin^{-1}(\frac{8}{17})$.

Answer: hypotenuse $= 17$; $\theta = \sin^{-1}(\frac{8}{17})$.

Example 8: Adjacent side from angle

Hypotenuse $= 18$, angle $= 28°$. Find the adjacent side.

Solution

Identify the ratio linking adjacent and hypotenuse: CAH.

$$\cos 28° = \frac{\text{adj}}{18}$$

Solve: $\text{adj} = 18 \cos 28°$.

Answer: $\text{adj} = 18 \cos 28°$.

Example 9: Inverse cosine

Adjacent $= 9$, hypotenuse $= 15$. Find θ.

Solution

Identify the ratio: CAH.

$$\cos \theta = \frac{9}{15} = \frac{3}{5}$$

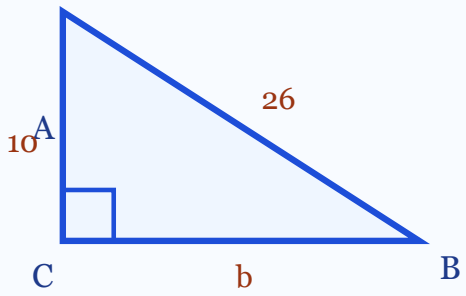
Solve using the inverse cosine: $\theta = \cos^{-1}(\frac{3}{5})$.

Answer: $\theta = \cos^{-1}(\frac{3}{5})$.

Example 10: Fully solve a triangle

A right triangle has leg 10 and hypotenuse 26. Solve the triangle completely (find the other leg and both acute angles).

Solution



One leg and the hypotenuse are known; find b first.

Find the other leg using the Pythagorean Theorem: $b^2 = 26^2 - 10^2 = 676 - 100 = 576$
$b = 24$.

Find one acute angle: $\theta = \sin^{-1}(\frac{10}{26}) = \sin^{-1}(\frac{5}{13})$.

Find the other acute angle: $\phi = \sin^{-1}(\frac{24}{26}) = \sin^{-1}(\frac{12}{13})$; note $\theta + \phi = 90°$ since the acute angles are complementary.

Answer: other leg $= 24$; angles $= \sin^{-1}(\frac{5}{13})$ and $\sin^{-1}(\frac{12}{13})$.

Your Turn — Show Your Work!

For each problem, solve it and verify your solution. Show all steps clearly.

Problem 1: Hypotenuse

Legs 5 and 12. Find c.

Answer: **Verification:**

Problem 2: Other acute angle

One acute angle is $41°$. Find the other.

Answer: **Verification:**

Problem 3: Opposite side

Hypotenuse $= 25$, angle $= 30°$. Find the opposite side (exact).

Answer: **Verification:**

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Problem 4: Inverse tangent

Opposite $= 7$, adjacent $= 24$. Express θ.

Answer: **Verification:**

Problem 5: Adjacent side

Hypotenuse $= 14$, angle $= 60°$. Find the adjacent side (exact).

Answer: **Verification:**

Problem 6: Hypotenuse

Adjacent $= 8$, angle $= 45°$. Find the hypotenuse (exact).

Answer: **Verification:**

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Problem 7: Inverse sine

Opposite $= 9$, hypotenuse $= 15$. Express θ.

Answer: **Verification:**

Problem 8: Hypotenuse

Legs 20 and 21. Find the hypotenuse.

Answer: **Verification:**

Problem 9: Opposite side

Adjacent $= 12$, $\tan\theta = \dfrac{5}{12}$. Find the opposite side.

Answer: **Verification:**

Problem 10: Other acute angle

Angle $A = 67°$ in a right triangle. Find angle B.

Answer: **Verification:**

Problem 11: Inverse cosine

Adjacent $= 8$, hypotenuse $= 17$. Express θ.

Answer: **Verification:**

Problem 12: Opposite side

Hypotenuse $= 30$, angle $= 45°$. Find the opposite side (exact).

Answer: **Verification:**

Problem 13: Challenge — fully solve

A right triangle has legs 10 and 24. Find the hypotenuse and both acute angles (as inverse-sine expressions).

Answer: **Verification:**

Problem 14: Challenge — angle and hypotenuse given

A right triangle has hypotenuse 34 and one acute angle $28°$. Find both legs (in terms of $\sin 28°$ and $\cos 28°$), then the other acute angle.

Answer: **Verification:**

Problem 15: Challenge — scaled special triangle

A right triangle is similar to a 3-4-5 triangle, scaled so its hypotenuse is 45. Find both legs, and express one acute angle as an inverse-sine ratio in lowest terms.

Answer: **Verification:**

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✓ Answer Key

1. 13
2. $49°$
3. 12.5
4. $\tan^{-1}(\frac{7}{24})$
5. 7
6. $8\sqrt{2}$
7. $\sin^{-1}(\frac{3}{5})$
8. 29
9. 5
10. $23°$
11. $\cos^{-1}(\frac{8}{17})$
12. $15\sqrt{2}$
13. Hypotenuse $= 26$; angles $= \sin^{-1}(\frac{5}{13})$ and $\sin^{-1}(\frac{12}{13})$
14. Opposite leg $= 34\sin 28°$; adjacent leg $= 34\cos 28°$; other angle $= 62°$
15. Legs $= 27$ and 36; one acute angle $= \sin^{-1}(\frac{3}{5})$

Great work — keep practising!